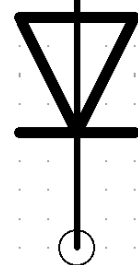


*How do people
design electronics??*

GND

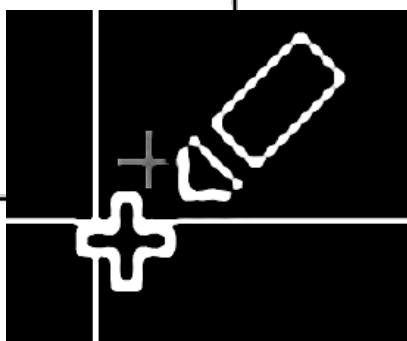
*A brief overview
for normies*

+12V

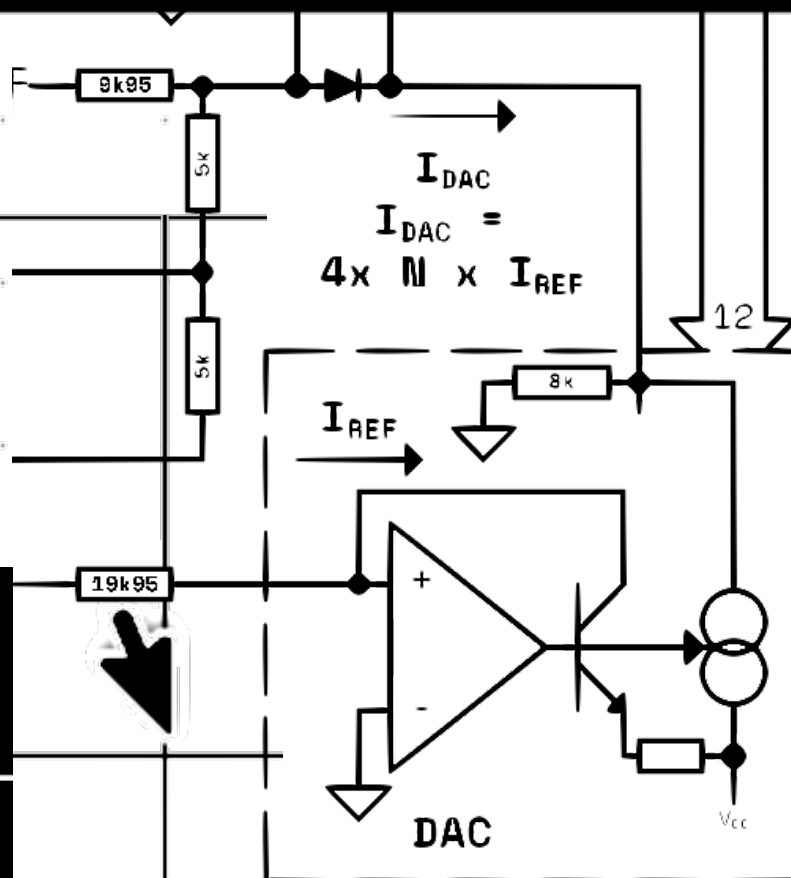


D1

BA283



V_{EE}



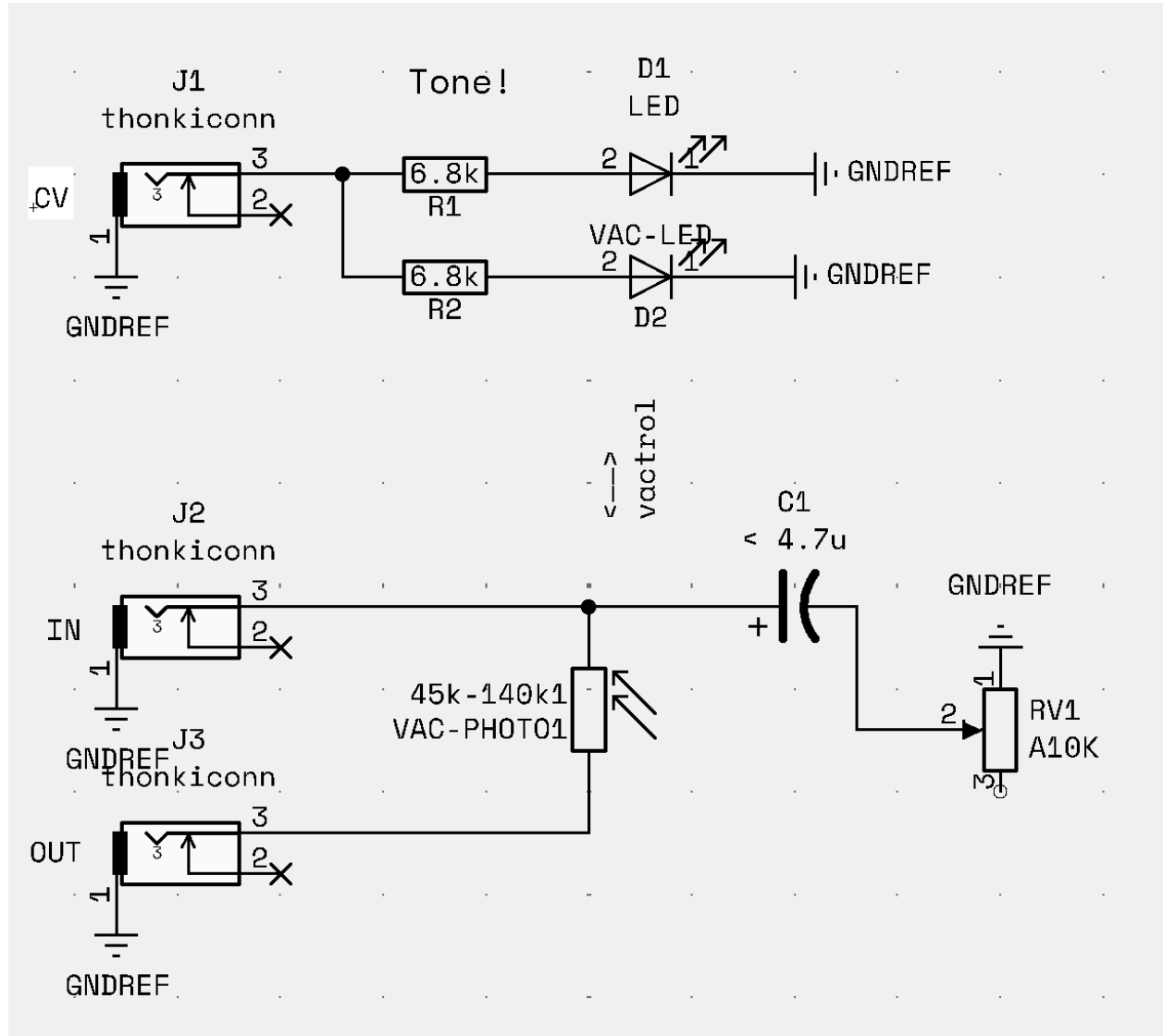
OUTPUT BUFFERS

I	DB4
B	DB5
B	DB6
L	DB7
E	
B	
N	DB8
I	DB9
B	DB10
L	DB11
E	
C	

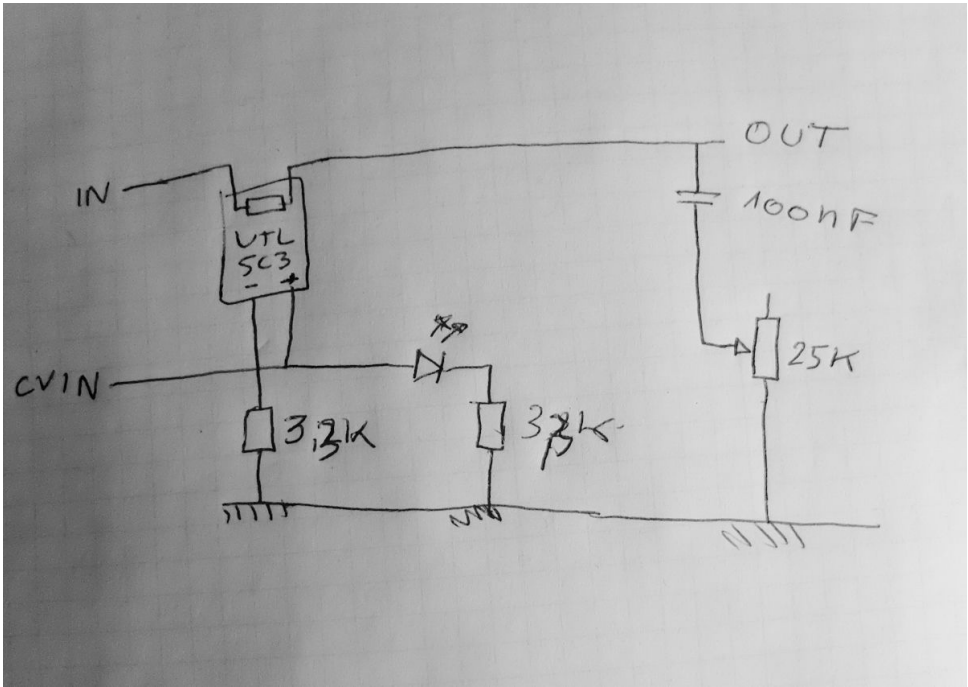
DC

* SCHEMATICS *

Electronic circuits work by connecting different “components” together with wires, creating all different types of functionality. Components are things like an LED, tiny microphone, or, a resistor, capacitor or transistor which limit, store, or amplify the flow of electricity in a circuit. A “Schematic” is a general map of the circuit. Think of it like a subway map; its purpose is to show the **logical connections** inside of the circuit, not show exact elements at exact scale/orientation. Each component has a generic symbol that represents it (alongside a name/number that describes specific model# or type). The symbols are connected by wires represented with lines. Some people draw these by hand, but usually they are made in software. Schematics help the engineer organize and document their own circuits, or understand/recreate someone else's. Designing the schematic is the first step in designing a circuit board, and is usually quite a technical and deep skill (you could also just use/copy an existing schematic)



*A simple schematic for a synthesizer ‘low pass gate’ (open source design called ‘two-tone’ by **bmoren**) being edited in the free and open source software **KiCad** (very popular and highly recommended).*



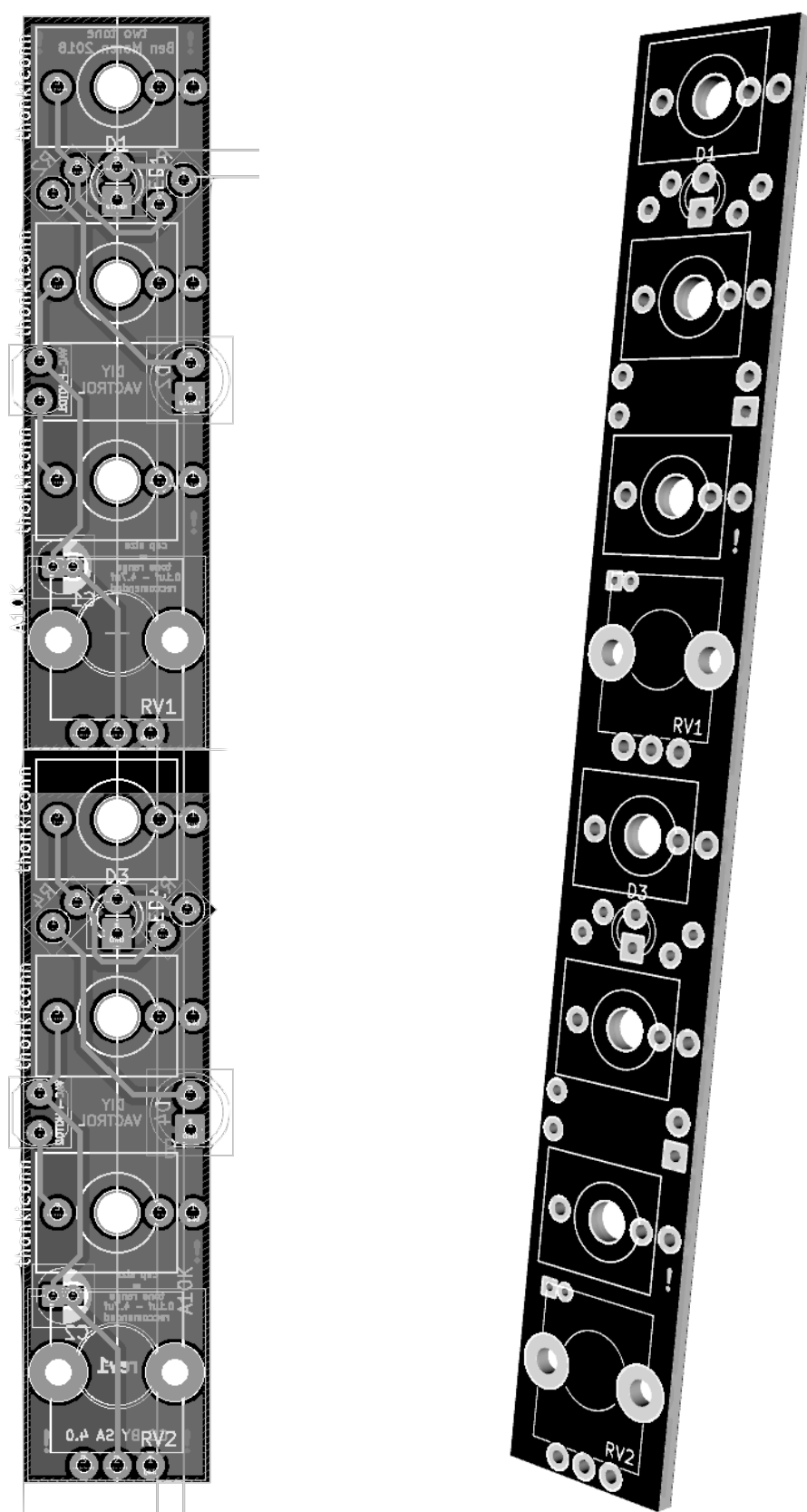
A simple hand draw schematic for a different low pass gate

** PCB Layout **

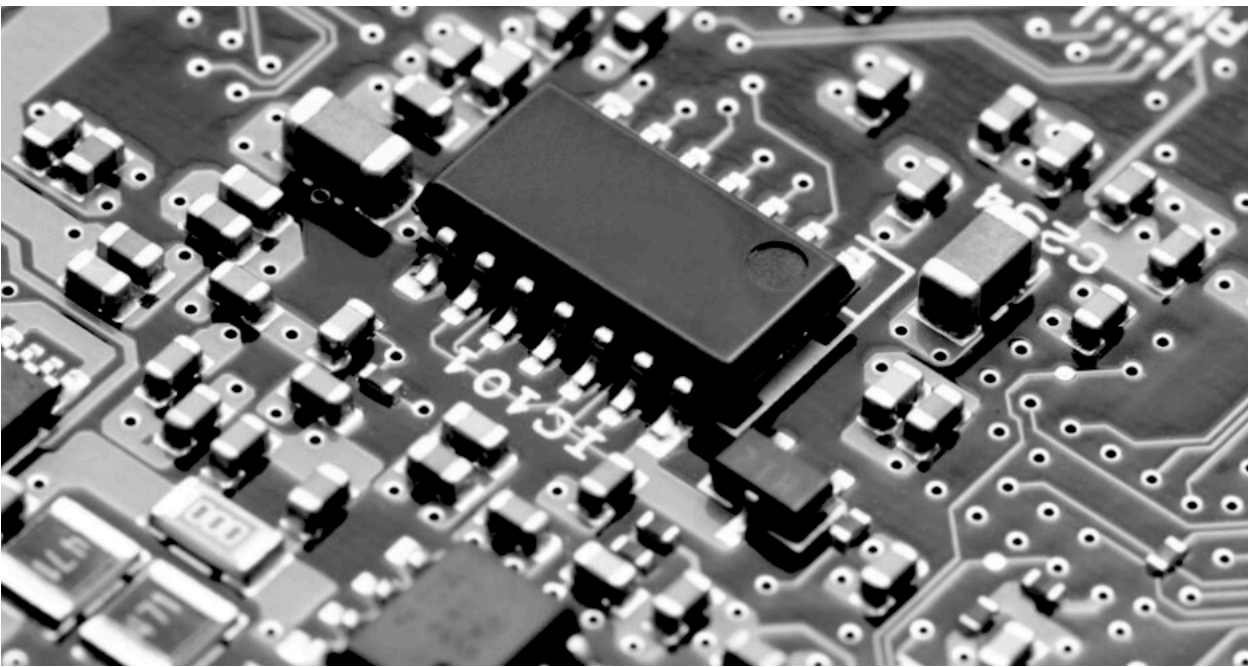
A PCB is a Printed Circuit Board (don't worry about 'printed' they're all printed). Once you have the logical map of the schematic, you can move into designing the actual layout of the circuit board, which is like a perfect, sub-millimeter accurate map of every component's placement, that you connect with accurately placed wires. Unlike in the schematic, in the PCB these components are a very specific make and model, and are true to real size.

A PCB is like a flat plastic-ish rectangle, that has small metal wires permanently stuck to it, which connect components which are held in place and electrically connected with little blobs of melted metal (called solder).

Making a PCB allows you to recreate the same circuit over and over much more easily and cleanly than remaking it from scratching each time. It's also much more organized having everything on one solid board

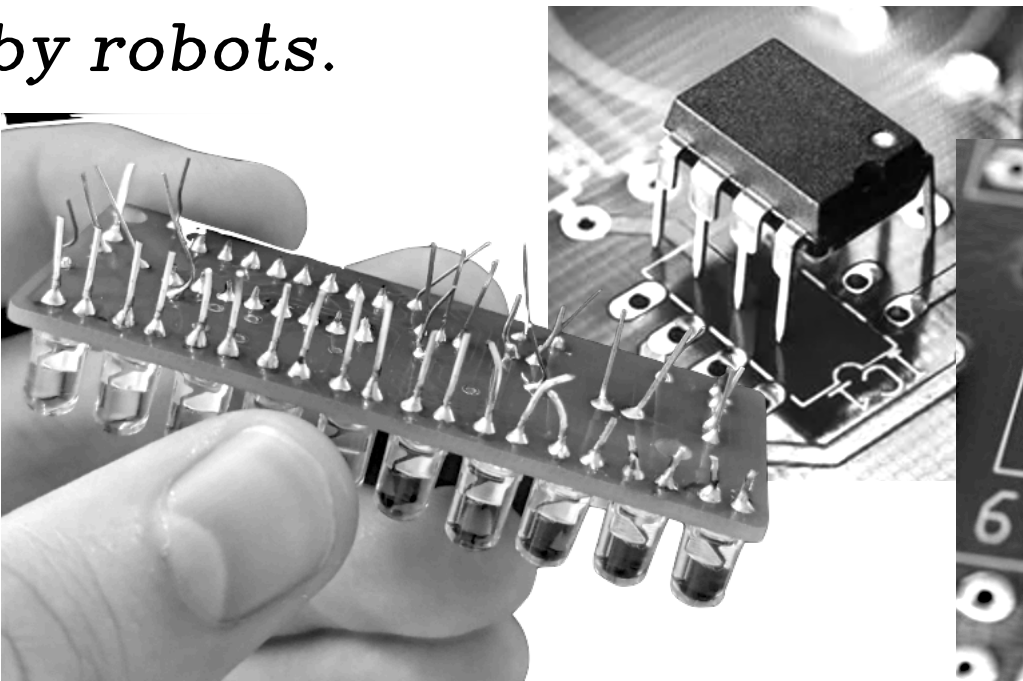


The *PCB* layout for the circuit board for the **two-tone**. Components are added by hand through the holes in the board and soldered on the back. The left is the KiCad *PCB* editor, the right is a render of the actual board from KiCad.

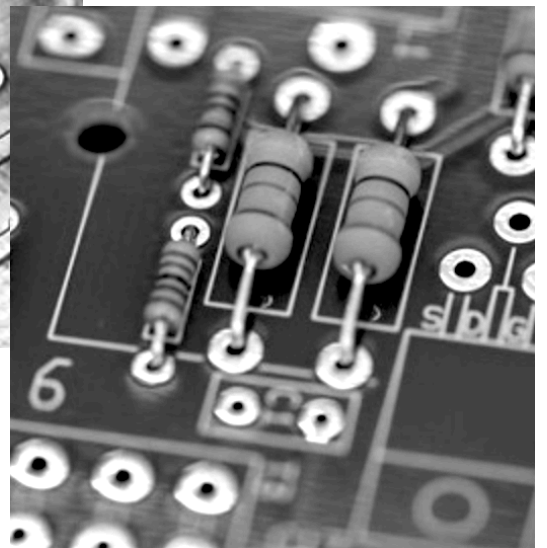


**Surface mount
(SMD / SMT)**

Most industrial electronics use 'surface mount' components which are small, sit right on top of the board, and are placed on the board and often soldered entirely by robots.



Thru Hole



Many older electronics or hobbyist boards primarily use **through hole** components which are placed into holes in the circuit board and soldered by hand by melting solder wire onto the part sticking through the back of the board. Lot of projects use A combination of both types

* Manufacturing *

Making a PCB is a complex process. Basically no-one does it themselves (you **could** do it in the same sense that you **could** build a car yourself). Large companies have dedicated factories (almost always in china) smaller companies or individuals place orders from on-demand factories (almost always in china) like JLCPCB, or PCBWAY (also AISLER and others in Germany). You generally export from your editing software of choice, to a format called **Gerber**, which is a generalized file that every factory takes (think of it like editing in word or google docs, but exporting to PDF for a printer). If you want to get components added and assembled by the factory its more complicated and you have to specify the component #s, placement, etc. When your boards arrive you add any extra components that need hand soldering, assemble any screws, faceplates, etc. And it's ready!



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By: JONAH SENZEL

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